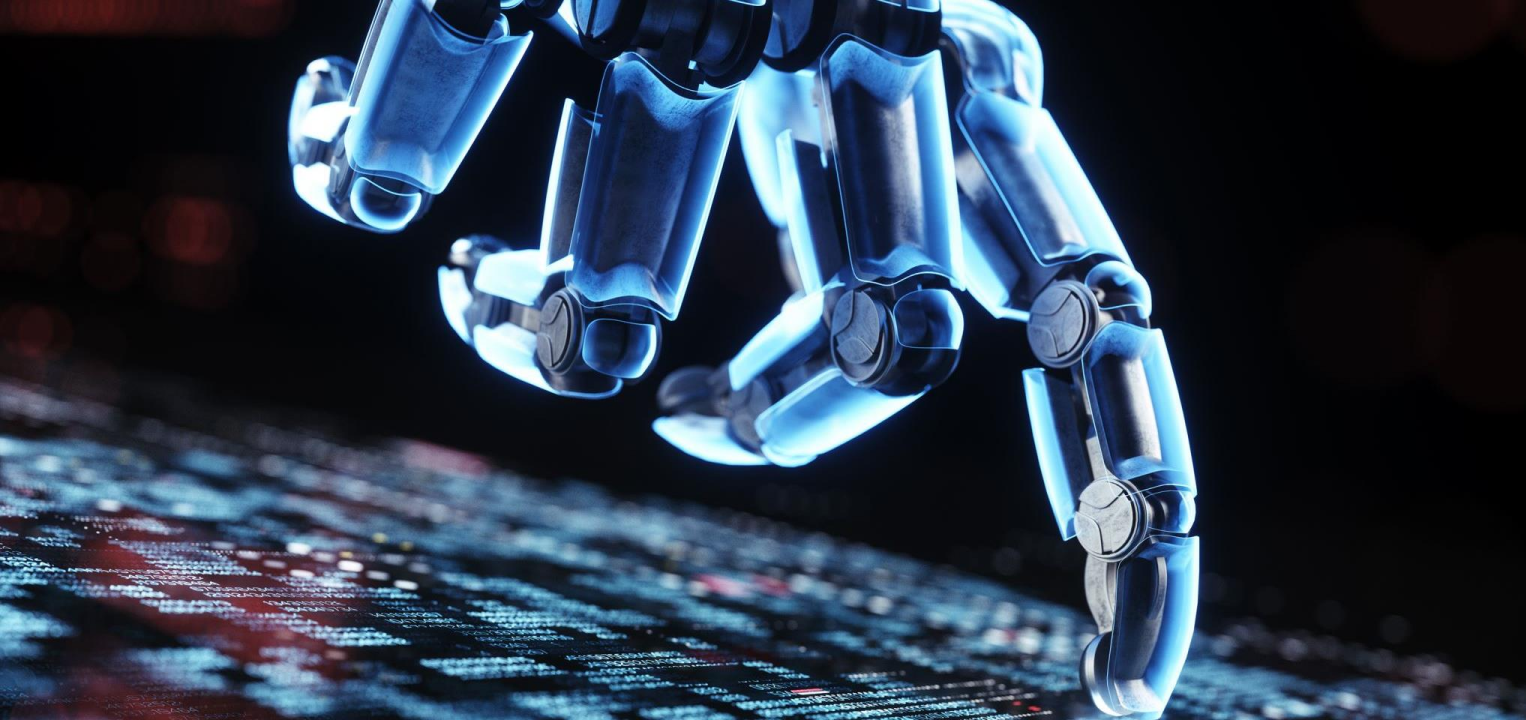




WORLD-MODEL-BASED
COGNITIVE ARCHITECTURE
FOR AUTONOMOUS
HUMANOID ROBOTS
USING
MULTIMODAL
PERCEPTION,
MEMORY, AND QUANTUM-
INSPIRED REASONING

PhD Research Proposal

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RESEARCH POSITIONING & SCIENTIFIC GAP

The proposal addresses a concrete unsolved problem in cognitive humanoid robotics — not just a system-building vision.

Research position

Embodied AI + world models
Long-term memory for tasks
Agentic executive for control

Scientific gap

- World models predict, but rarely update with persistent episodic memory
- Reasoning under uncertainty remains weak in unseen environments
- Humanoid deployment needs a unified cognitive architecture

PhD target

Develop and validate a cognitive architecture linking perception, memory, prediction, reasoning, planning, and action.

Core thesis: move humanoid robots from reactive execution toward cognitive autonomous agents that recall, predict, and decide under uncertainty.

LITERATURE REVIEW SHARPENS THE RESEARCH GAP

The proposal is strongest when framed as an integration problem: memory, world-model updating, agentic control, and uncertainty-aware reasoning.

Research trend

- World models support prediction and planning
- Embodied AI connects perception to action
- Agentic systems add planning, memory, and feedback

Unsolved gap

- Long-term memory is not yet tightly coupled with humanoid world models
- Robots still struggle to update beliefs in unseen environments
- Evaluation must test adaptation, not only task success

Proposal response

- Define explicit research questions
- Add agentic executive and validation metrics
- Compare against planning and memory baselines

Core research claim: humanoid robots need persistent memory, adaptive world models, and uncertainty-aware reasoning to move beyond reactive control.

EXPLICIT RESEARCH GAP

Current embodied AI connects perception and action, but still lacks persistent cognition for dynamic, unseen environments.

Observed limitations

- Limited long-term episodic memory
- Poor reasoning under uncertainty
- Weak world-model updating
- Fragmented cognitive pipeline

World-model gap

- Prediction is improving
- Planning is improving
- Memory integration remains weak
- Humanoid deployment remains rare

Scientific problem

- Connect perception
- Recall past experiences
- Imagine future outcomes
- Decide under uncertainty

Research gap: world models improve prediction and planning, but are rarely integrated with persistent memory and uncertainty-aware reasoning inside humanoid robots.

RESEARCH QUESTIONS

RQ1

How can multimodal perception become a unified latent world representation?

RQ2

How can episodic and semantic memory improve long-horizon humanoid tasks?

RQ3

How can a world model predict future environmental states for planning?

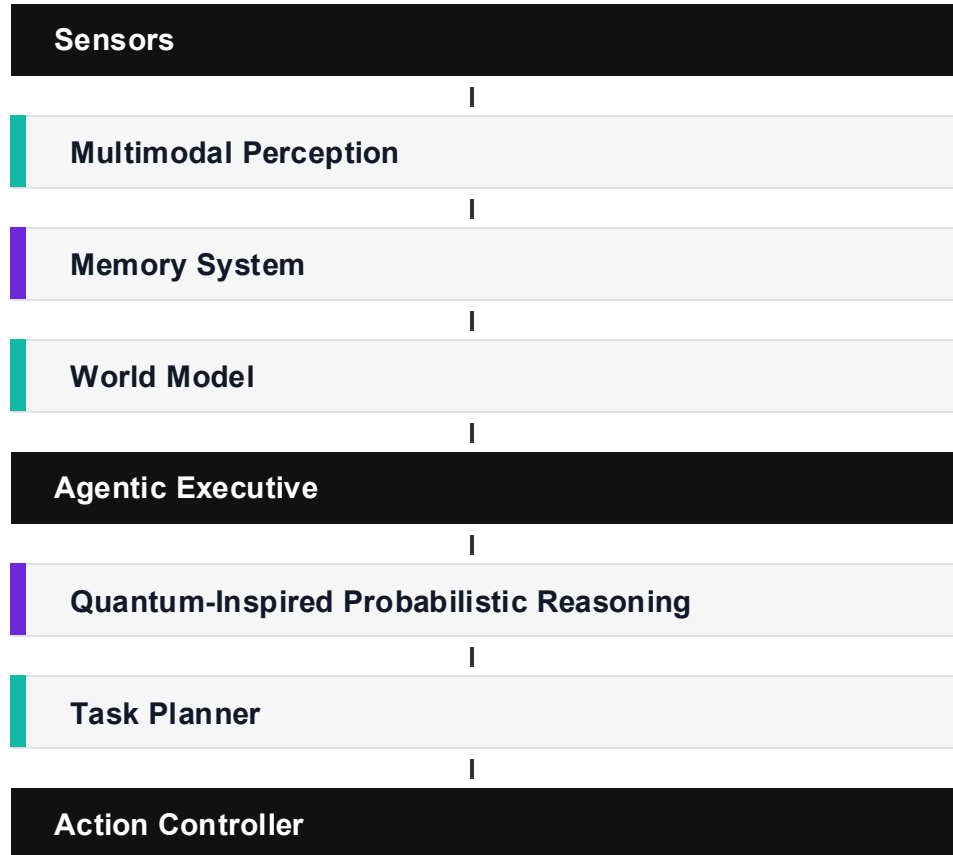
RQ4

Can quantum-inspired probabilistic reasoning improve decisions under uncertainty?

RQ5

How can these modules form a scalable cognitive architecture for humanoid robots?

UPDATED COGNITIVE ARCHITECTURE



Key innovation: cognitive control loop

Agentic Executive Layer

Maintains goals, decomposes tasks, monitors progress, requests memory/world-model updates, and routes uncertainty to the reasoning engine.

Memory-Augmented World Model

Connects episodic, semantic, and working memory to predictive state estimation so the robot reasons from experience.

Embodied Action Stack

ROS2 planning and action control translate cognitive decisions into navigation, manipulation, and human-robot interaction.

VALIDATION METHODOLOGY

Environment

- ROS2
- Gazebo
- Isaac Sim
- Simulation-to-real pipeline

Tasks

- Navigation
- Object search
- Human-robot interaction
- Long-horizon planning
- Unseen scenario adaptation

Metric	What it tests
Task success rate	End-to-end execution
Memory recall accuracy	Episodic / semantic retrieval
Planning horizon length	Future-state prediction depth
Adaptation time	World-model updating speed
Decision confidence	Uncertainty-aware action choice

Validation claim: measure task success, memory, planning depth, adaptation, decision confidence, and ablation baselines.

EXPECTED CONTRIBUTIONS

1

Novel cognitive architecture

Integrates perception, memory, world model, agentic executive, decision engine, planner, and controller.

2

Memory-augmented world model

Connects episodic, semantic, and working memory with predictive latent-state modeling for embodied agents.

3

Quantum-inspired reasoning framework

Provides an uncertainty-aware probabilistic decision layer for belief-state optimization and action selection.

4

Simulation-to-real deployment pipeline

Implements the architecture through ROS2, Gazebo, Isaac Sim, and progressive humanoid deployment tests.

5

Benchmark and evaluation framework

Defines task success, memory recall, planning horizon, adaptation time, and decision confidence metrics.

FUTURE IMPACT & DEFENSE-READY THESIS

Core thesis: enable humanoid robots to build world representations, recall past experiences, imagine future outcomes, and make uncertainty-aware decisions.

Application domains

- Service and assistive robots
- Healthcare and eldercare robotics
- Industrial automation
- Human-robot collaboration
- Smart manufacturing

“The proposed architecture enables humanoid robots to build internal world representations, recall past experiences, imagine future outcomes, and make uncertainty-aware decisions, thereby moving from reactive robots toward cognitive autonomous agents.”